Thermal boundary resistance of bonded semiconductor interfaces

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***Abstract*—Thermal interfaces between heterogeneous integrated semiconductors have emerged as a critical bottleneck for effective thermal dissipation. This summary reviews recent advancements, highlighting a significant enhancement in thermal boundary conductance (TBC) following annealing, alongside the development of a novel TBC measurement technique that offers a high signal-to-noise ratio and enables fast TBC mapping.**

I. Introduction And Background

Thermal management is a critical challenge for today's high-frequency and high-power devices. The heterogeneous integration of high thermal conductivity materials, such as SiC and diamond, has been proposed as a promising solution, with TBC playing a pivotal role. High TBC interfaces are, therefore, essential for effective thermal management. This summary focuses on the fabrication of high TBC interfaces between 3C-SiC and diamond, which exhibit a significant TBC improvement after annealing. Additionally, an effective TBC mapping technique, known as liquid immersion probe beam deflection (LI-PBD), is introduced.

II. Results

SiC possesses outstanding electrical and thermal properties, with 3C-SiC having the second highest thermal conductivity (κ) among large crystals. Constructing a GaN/3C-SiC/diamond structure significantly alleviates the thermal challenges in GaN devices[1]. While high TBC has been achieved at the SiC/GaN interface[2], there is now a growing demand for highly conductive 3C-SiC/diamond interfaces to bypass the challenge of direct integration of GaN with diamond.

Using surface-activated bonding (SAB), a thin yet highly thermally conductive 3C-SiC film was bonded onto a diamond substrate[3]. A 10 nm Si layer was pre-deposited onto the diamond surface. After bonding, an amorphous interlayer initially formed, but as the annealing temperature increased, it gradually repaired. Notably, at an annealing temperature of 1100 °C, the TBC of the interface experienced a remarkable increase of nearly 300%, reaching 150 MW/m²K—a record high among diamond-related interfaces. This boost in TBC is attributed to the formation of a 3C-SiC layer during annealing. The bonding quality uniformity was evaluated through TDTR mapping. Experimental results of 3C-SiC thin-film κ and SiC/diamond TBC are shown in Fig. 1, alongside comparisons to other reported values.

LI-PBD is a novel thermal characterization technique. Unlike traditional thermoreflectance methods, which measure intensity changes in the probe beam, LI-PBD measures its spatial displacement caused by thermal expansion. The pump and probe beams are offset by a few micrometers, and the probe is deflected by surface displacement induced by the pump beam. The presence of liquid in the optical path amplifies this effect and improves the signal-to-noise ratio (SNR). By fitting the in-phase and out-of-phase signals, the TBC can be extracted using a thermal model. The experimental setup and mapping results are shown in Fig. 2 [4].

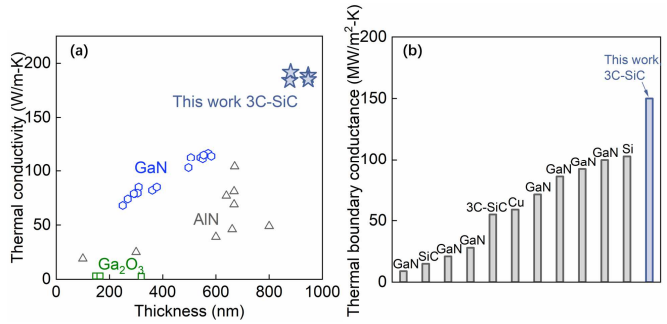


Fig. 1. Record-high thermal conductivity of 3C-SiC thin films and TBC of 3C-SiC/diamond interface. (a) Comparison of thermal conductivity of thin films. (b) Comparison of TBC of diamond-related interfaces. Adapted from [3].

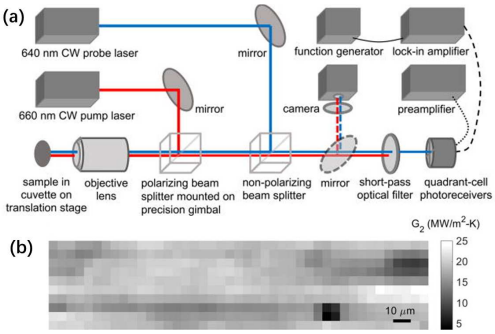


Fig. 2. LI-PBD measurement. (a) The setup of LI-PBD. (b) The mapping results of the TBC of a buried interface of SiC/diamond. Adapted from [4].

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